Aashna Sibal

PSet 1 DUE 1/31

1) a) The best 1-leaf decision tree always makes one prediction, the majority class[1]. This would be incorrect in cases where $X_1 = X_2 = X_3 = 0$. These would be mistakes, since the true label is 0. There are 2^{n-3} possible vector combinations where this mistake would occur, since the first three X_i values are fixed as 0, leaving 2 options for the remaining $n-3$ features (0 or 1).

$$\frac{2^{n-3}}{2^n} = \frac{\text{mistakes}}{\text{total}} = 2^{n-3-n} = 2^{-3} = \frac{1}{8}$$

$\frac{1}{8}$ of the total 2^n training examples will result in a mistake. Or, 2^{n-3} mistakes.

b) NO - no matter which variable we split on, the proportion of ones in each leaf will be greater than the proportion of 0s. Therefore, each leaf would predict 1 - just like in the 1-leaf decision tree. Even if we split on X_i where $i = 1, 2$ or 3, 1 leaf would be all 1's ($X_i = 1$), and the other would have $\frac{3}{4}$ 1's ($X_i = 0$). With the same predictions, we maintain the same # of mistakes.

c) $H(X) = -\left[\left(\frac{1}{8}\right)\log_2\left(\frac{1}{8}\right) + \left(\frac{7}{8}\right)\log_2\left(\frac{7}{8}\right)\right] = 0.544$ bits

d) Yes, splitting on X_1, X_2, or X_3 would reduce the entropy.

$P(X_i = 0) = \frac{1}{2}$ $P(X_i = 1) = \frac{1}{2}$

$P(Y=0) = \frac{1}{4}$ $P(Y=1) = \frac{3}{4}$ y must be 1, entropy is 0

$-\frac{1}{2}\left(\frac{1}{4}\log_2\frac{1}{4} + \frac{3}{4}\log_2\frac{3}{4}\right)$ + $\frac{1}{2}(0) = 0.406$ bits

$0.544 - 0.406 = 0.138$ bits. With a resulting conditional entropy of 0.406, we reduced the entropy by 0.138 bits.

2) a) $B(q) = -q\log q - (1-q)\log(1-q)$ $\quad u=-q \quad v=\log q \quad j=1-q \quad K=\log(1-q)$

$\frac{\partial B}{\partial q} = (u'v+v'u) - (j'K + K'j)$ $\quad u'=-1 \quad v'=\frac{1}{q\ln 2} \quad j'=-1 \quad K'=\frac{-1}{(1-q)\ln 2}$

$\frac{\partial B}{\partial q} = -\log q - \frac{q}{q\ln 2} - \left(-\log(1-q) + \frac{-(1-q)}{(1-q)\ln 2}\right)$

$\frac{\partial B}{\partial q} = -\log q - \cancel{\frac{1}{\ln 2}} + \log(1-q) + \cancel{\frac{1}{\ln 2}} = \log\left(\frac{1-q}{q}\right) = \boxed{\log(1-q) - \log(q)}$

$\frac{\partial^2 B}{\partial q^2} = \frac{-1}{(\ln 2)(1-q)} - \frac{1}{q\ln 2} = \boxed{\frac{-1}{\ln 2}\left(\frac{1}{1-q} + \frac{1}{q}\right)}$

$\downarrow \frac{\partial^2 B}{\partial q^2}$

since this will always be negative when $0 \le q \le 1$, H is a concave function.

Maxima: $\frac{\partial B}{\partial q} = 0$

$\log\left(\frac{1-q}{q}\right) = 0$

$\frac{1-q}{q} = 1$

$1-q = q$

$1 = 2q$

$q = \frac{1}{2}$ ← maximum occurs at

Since the maximum is $\frac{1}{2}$, plug into B.

$B(\frac{1}{2}) = -\frac{1}{2}\log\frac{1}{2} - (\frac{1}{2}\log\frac{1}{2}) = \boxed{1}$

Thus, $H(S) \le 1$. Since fn concave, find minimums using bounds.

$B(0) = 0 - 1(\log 1) = 0$ $\quad$ Thus, $0 \le H(S) \le 1$. □

$B(1) = -1(\log 1) - 0 = 0$

$\updownarrow$

$p=n \quad B(\frac{p}{p+p}) = B(\frac{1}{2})$

$H(S) = B(\frac{1}{2}) = -\frac{1}{2}\log\frac{1}{2} - (\frac{1}{2})\log\frac{1}{2} = -\frac{1}{2}(-1) - (\frac{1}{2}\cdot -1) = \frac{1}{2} + \frac{1}{2} = 1$ □

$H(S) = 1$ when $p = n$.

b) $H(S_K) = B\left(\frac{p_K}{p_K + n_K}\right)$

$H(S) - \sum_{i=1}^{K} \frac{|S_i|}{|S|} H(S_i)$

gain = $H[Y] - H[Y|X]$

where $H(S_i)$ refers to the entropy of the subset i, and $\frac{|S_i|}{|S|}$ refers to ~~probability~~ weight.

In $B(q)$, the entropy function, it relies on the proportion of pos and neg examples. Since the proportion of p and n to total examples is the same for every subset (as stated in the problem), each subset has same entropy as the whole set S, $H(S_i) = H(S)$.

$$H(S) - \sum_{i=1}^{K} \frac{|S_i|}{|S|} H(S) = H(S)\left[1 - \sum_{i=1}^{K} \frac{|S_i|}{|S|}\right] = H(S)[1-1] = \boxed{0} \quad \square$$

Info gain = 0.

↑ must sum to 1, to ensure the subsets includes all examples in the set. probabilities add up to 1.

3) a) K=1 would minimize the training set error, since each point's 1 nearest neighbor would be itself. ~~Each~~ Each predicted label would match the the actual label with perfect accuracy, resulting in a training error of 0.

Training set error is not a reasonable estimate of test set error since the model could be overfitting. Especially for this value of K, the model is just memorizing the point, not actually learning anything about the general trends/patterns. A model could have very low training error but high testing set error, since it did not learn general trends and patterns that would help it make correct predictions on new, unseen data. Overall, this would result in a model that is not very good at generalizing to new data.

LOOCV

b) K=5 or K=7 would minimize the training set error. This would result in error of 4/14. The four odd ones out (2 * on left half and 2 O on right half) would be misclassified. If K=7 or K=5, everything on the left is O and everything on the right is * predicted. Anything larger or smaller would result in a higher error.

Cross-validation is a better measure of test-set performance because it better captures how the model performs on unseen data. In each fold, we are separating some of the data to act as the unseen data. Averaging these scores (like cross-validation does) gives a general estimate of what the model would perform like on new unseen data. If the model performs well on these different combinations of training data/unseen data, it likely has a good ability to generalize to new data.

c) lowest K: K = 1

We would have LOOCV error of 10/14. The 5 points centered in the left half and those in the right half get misclassified since their nearest neighbor is the opposing class.

highest K: K = 13 (since we must leave one out)

Each point would be misclassified, since the majority class would be it's opposing class, since you must leave that point out.

LOOCV error = 14/14

Having a very small value of K could result in overfitting to the data and thus a lack of the ability to generalize to new unseen data. Having a very large value of K like 13 would result in underfitting, since each point would be classified as what the majority class is (after removing a point) rather than actually capturing trends. Again, this results in poor performance on new data, and a model that is too simple. Even a large K in general, while it may get some points correct, could be underfitting, and too simple.

CM146, Winter 2025
Problem Set 1: Decision Trees, Nearest Neighbors
Due Jan 31, 2025 at 11:59 pm PST

1 Problem 4

(a) Problem 4a

Solution:

Workclass: >50k seemed to be much more evenly distributed throughout the different working classes, while <50k was very concentrated at workclass = 1. However, both peaked at workclass = 1.

Education: Again, both classes peak at 2 education. <50k had a more widespread distribution, since it had 10 bars present, representing 10 education labels, while >50k only had 6 bars present.

Martial Status: Making more than 50k was most common for those in marital status 1. Making less than 50k was most common for those in martial status 3.

Occupation: >50k peaks at around 5 education and has a normal distribution. <50k peaks at 3 education and has a right-skewed distribution.

Relationship: >50k peaks at 3, while <50k peaks at 4. Both have a right-skewed distribution.

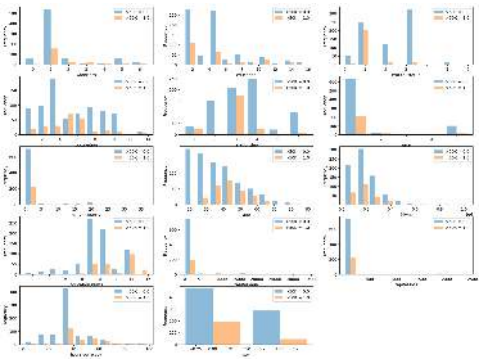


Figure 1: Histograms

Race: There is very little data for both classes in groups race = 2, 3, or 4. Both have the most data in race = 1.

Native Country: Most of the data for both classes is concentrated around native country = 1. >50k is only in this country, while <50k has bars through 10-20 as well.

Age: Both classes have a right-skewed distribution, but much more so for <50k. The most common age for <50k is 20, while the most common age for >50k is around 45.

Fnlwgt: Both classes are again right skewed. Both classes also peak at 0.2.

Education Nums: Both classes are left skewed. However, <50k peaks at 9, while >50k peaks at 14.

Capital Gain: <50k only has datapoints under or equal to 5000, but >50k has data points ranging from 0 to 16000. Still, both peak at between 0-5000.

Capital Loss: There is very little data for both classes beyond 250 capital loss. Both peak at under 250.

Hours per Week: <50k has a widespread distribution ranging from 0-80. >50k has points only ranging from 30-80.

Sex: Both sexes have more datapoints in the <50k category. The discrepancy across classes is a little larger for the sex = 0 (difference of around 300, while for sex = 1 it is around 250).

(b) Problem 4b

Solution: My training error: 0.385

(c) Problem 4c

Solution: My training error: 0.000

(d) Problem 4d

Solution:

k = 3 training error: 0.153

k = 5 training error: 0.195

k = 7 training error: 0.213

(e) Problem 4e

Solution:

Majority Vote Classifier: Train Error: 0.239, Test Error: 0.239, F1 Score: 0.760

Random Classifier: Train Error: 0.371, Test Error: 0.373, F1 Score: 0.627

Decision Tree Classifier: Train Error: 0.0, Test Error: 0.211, F1 Score: 0.789

KNN Classifier: Train Error: 0.202, Test Error: 0.259, F1 Score: 0.741

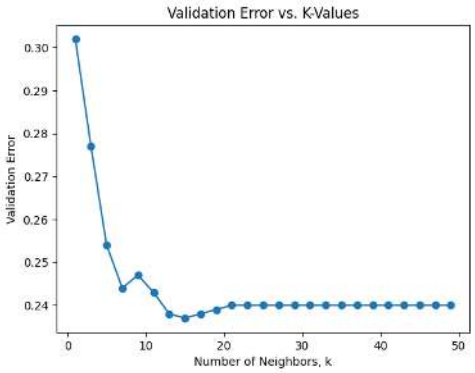


Figure 2: Validation Error vs K-Values

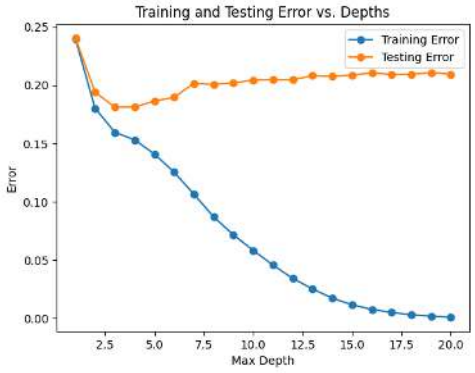


Figure 3: Error vs Depth

(f) Problem 4f

Solution: We see a significant drop in validation error as k goes from 1-7, while it is consistent after 21. The best k-value would be k = 15, since that is when the validation error is the lowest in our graph (Figure 2).

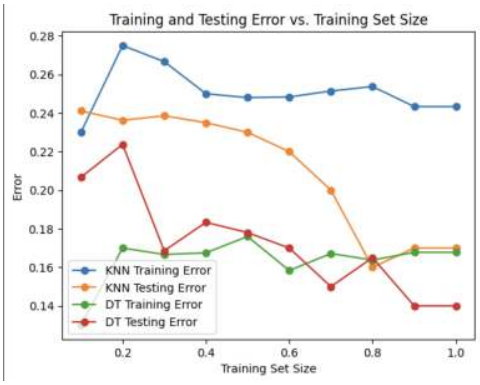


Figure 4: Learning Curves

(g) Problem 4g

Solution: The best depth limit is 3, since this is the minimum testing error value we see on the graph (Figure 3). Also, it is still fairly close to the training error. We definitely see overfitting towards the end, since the training error is approaching 0 while the testing error remains at its highest. Therefore, it wold not be smart to use the very large depth limits we tested.

(h) Problem 4h

Solution: As the training set size increases, the training set error goes down for KNN and not as much for DT. As the training size set increases, testing error for both classifiers has a general downwards trend. (Figure 4)

(i) Problem 4i

Solution: Parts a-c remained the same. When looking at part d, the training error went down about 0.05 for each k-neighbor values. MajorityVoteClassifier, Random Classifier, and Decision Tree Classifier returned the same train and test error and F1 score. There was a different with KNN classifier, with around a 0.07 decrease in train error, 0.05 decrease in test error, and 0.05 increase in F1 score.

The best k-value is now k = 27, and the best depth is still 3.

As for the learning curves, the training and testing errors are much closer now in general trends, as shown by the gap between lines being smaller and overlap across the lines.